

Understanding Section 2.7: Upper-Confidence-Bound Action Selection

Multi-Armed Bandits Study Notes

Problem Statement

In the multi-armed bandit setting, exploration is needed because the current action-value estimates may be inaccurate. A greedy action is the action that currently looks best, but some non-greedy action may actually have a higher true value. The standard ε -greedy method forces exploration by randomly selecting non-greedy actions with probability ε , but this exploration is indiscriminate.

Section 2.7 introduces **Upper-Confidence-Bound (UCB) action selection**, which explores more intelligently by considering both:

- (1) the current estimated value of an action, and
- (2) the uncertainty in that estimate.

The UCB action selection rule is

$$A_t = \arg \max_a \left[Q_t(a) + c \sqrt{\frac{\ln t}{N_t(a)}} \right], \quad (1)$$

where $Q_t(a)$ is the estimated value of action a at time t , $N_t(a)$ is the number of times action a has been selected before time t , $\ln t$ is the natural logarithm of t , and $c > 0$ controls the degree of exploration.

If $N_t(a) = 0$, then action a is considered a maximizing action. In practice, this means UCB will try every action at least once.

Core Idea

UCB chooses the action with the largest optimistic estimate:

$$\text{UCB score}(a) = \underbrace{Q_t(a)}_{\text{exploitation}} + \underbrace{c \sqrt{\frac{\ln t}{N_t(a)}}}_{\text{exploration bonus}}. \quad (2)$$

The first term, $Q_t(a)$, favors actions that currently look good. This is the exploitation term.

The second term,

$$c\sqrt{\frac{\ln t}{N_t(a)}}, \quad (3)$$

favors actions that have high uncertainty. This is the exploration bonus. The fewer times an action has been selected, the larger this bonus becomes.

Therefore, UCB does not explore randomly. It explores actions that are still uncertain and may plausibly be optimal.

Meaning of Each Symbol

Symbol	Meaning
A_t	action selected at time step t
$Q_t(a)$	current estimate of the value of action a
$N_t(a)$	number of times action a has been selected before time t
$\ln t$	natural logarithm of the current time step t
c	positive parameter controlling exploration strength

Table 1: Notation used in UCB action selection.

Why the Exploration Bonus Works

The exploration bonus is

$$c\sqrt{\frac{\ln t}{N_t(a)}}. \quad (4)$$

Consider the denominator $N_t(a)$. If action a has been selected many times, then $N_t(a)$ is large, and the exploration bonus becomes small. This means UCB becomes more confident about that action and explores it less.

If action a has been selected only a few times, then $N_t(a)$ is small, and the exploration bonus becomes large. This means UCB treats the action as uncertain and gives it more opportunities to be tested.

The numerator $\ln t$ grows slowly as time passes. If an action has not been selected for a long time, then t increases while $N_t(a)$ does not. As a result, the exploration bonus for that action slowly increases. This prevents UCB from permanently ignoring an action.

Because the logarithm grows slowly, the exploration pressure increases gradually rather than aggressively. For example,

$$\ln(10) \approx 2.30, \quad \ln(1000) \approx 6.91. \quad (5)$$

Thus, UCB keeps exploration alive, but it does so in a controlled way.

What Happens When $N_t(a) = 0$?

If an action has never been selected, then $N_t(a) = 0$. The expression

$$\sqrt{\frac{\ln t}{N_t(a)}} \quad (6)$$

would involve division by zero. To avoid this, the rule defines every untried action as a maximizing action.

In practical terms, this means UCB tries each action at least once before relying fully on the formula. In a k -armed bandit, this causes UCB to spend the first k steps sampling untried actions.

This explains why UCB may perform worse than ε -greedy at the very beginning: it is forced to try actions whose rewards may be poor.

Comparison with ε -Greedy

The ε -greedy method usually chooses the greedy action, but with probability ε it selects a random action. Its exploration is simple but indiscriminate.

UCB is more selective. It prefers actions that either have high estimated value or high uncertainty.

Method	Exploration Mechanism	Main Weakness
ε -greedy	random exploration	may waste trials on clearly poor actions
UCB	uncertainty-guided exploration	harder to extend beyond simple bandits

Table 2: Comparison between ε -greedy and UCB action selection.

The difference can be summarized as follows:

$$\varepsilon\text{-greedy : random exploration,} \quad (7)$$

$$\text{UCB : optimistic, uncertainty-guided exploration.} \quad (8)$$

Understanding the Figure

Figure 1 compares UCB with $c = 2$ against ε -greedy with $\varepsilon = 0.1$ on the 10-armed testbed.

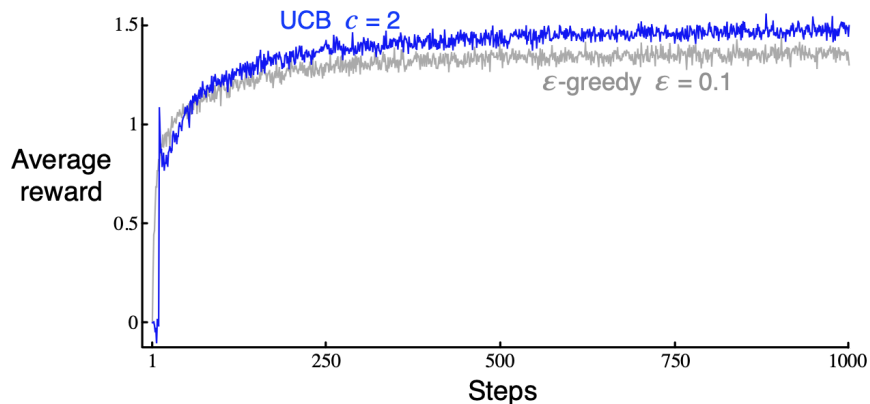


Figure 2.4: Average performance of UCB action selection on the 10-armed testbed. As shown, UCB generally performs better than ε -greedy action selection, except in the first k steps, when it selects randomly among the as-yet-untried actions.

Figure 1: Average performance of UCB action selection compared with ε -greedy action selection on the 10-armed testbed.

At the beginning, UCB performs worse because it tries all unselected actions. Some of these actions are bad, so the early average reward can be low.

After the initial exploration phase, UCB improves quickly. It has gathered enough information to distinguish promising actions from weak ones, and its uncertainty bonus helps it continue exploring only when it is useful.

Over time, UCB achieves a higher average reward than ε -greedy because it explores more efficiently.

Simple Numerical Example

Suppose there are three actions with the following estimates and selection counts:

Action	Estimated value $Q_t(a)$	Times selected $N_t(a)$
A	1.2	100
B	1.1	5
C	0.8	2

Table 3: Example action-value estimates and counts.

A greedy method would choose action A because it has the highest estimated value.

However, UCB may choose B or C if their uncertainty bonuses are large enough. Action B has a slightly lower estimate than A, but it has been tried only 5 times. Action C has an even lower

estimate, but it has been tried only 2 times. UCB balances the estimated value against uncertainty.

Thus, UCB does not simply choose the action that currently looks best. It chooses the action with the best optimistic upper estimate.

Why UCB Is Called Optimistic

UCB follows the principle of **optimism in the face of uncertainty**. For each action, it forms an optimistic estimate:

$$\text{optimistic value} = \text{estimated value} + \text{uncertainty bonus}. \quad (9)$$

If an action is uncertain, UCB gives it the benefit of the doubt. But after the action has been sampled many times, its uncertainty bonus decreases. If the action is truly bad, UCB will eventually stop selecting it often.

This is the key benefit of UCB: it encourages useful exploration without relying on completely random action selection.

Why UCB Is Harder Beyond Bandits

In the bandit setting, there is only one set of actions, so we can maintain a simple count $N_t(a)$ for each action.

In general reinforcement learning, action values depend on states:

$$Q(s, a). \quad (10)$$

Now uncertainty depends on both state and action. In large state spaces, many states may be visited rarely, and with function approximation such as neural networks, it becomes difficult to define a simple and reliable uncertainty bonus.

This is why UCB is elegant and effective for bandit problems, but less practical as a direct method for large reinforcement learning problems with many states or function approximation.

Key Takeaway

UCB improves over ϵ -greedy because it explores intelligently. The rule can be read as:

Choose the action with the highest current estimate plus an exploration bonus for uncertainty.

Equivalently,

$$\text{UCB score} = \text{current estimated value} + \text{uncertainty bonus.} \quad (11)$$

This allows UCB to focus exploration on actions that are still uncertain and potentially optimal, instead of wasting exploration uniformly across all non-greedy actions.